Masters Thesis: Computation Modelling of Buoyancy-driven Flow

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Technical Abstract

Katabatic wind is a striking phenomenon in long valleys. It is a result of cooler air next to the mountain surface being driven down by buoyancy. Because it involves the interaction of a viscous boundary layer and a thermal boundary layer, it is interesting to study and relevant to the engineering course.

Prandtl has devised a model of the boundary layer flow for an open ended valley, shown in Fig. 1 below. However, no current work exists that matches the core flow to that in the boundary layer. Even less well understood is the more complex case where the valley is blocked on the ends and the flow has to recirculate in the 2-D cross-sectional plane (Fig. 2). This project examines recirculating flow in two separate parts, the core flow and the boundary layer.



Fig. 1. An Open Valley

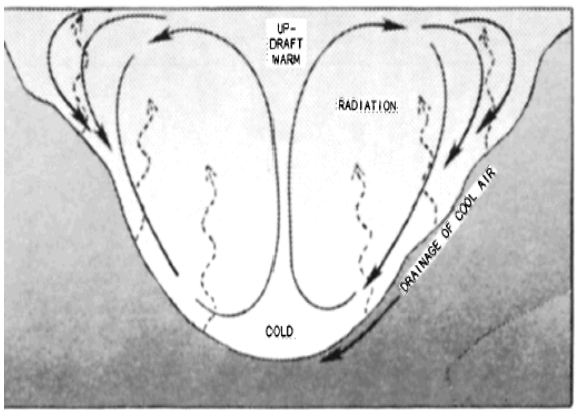


Fig. 2. A Blocked Valley

Inside the core under steady state, the temperature is found to only depend on the vertical distance. All core streamlines have to pass through the boundary layer to dissipate energy gained from buoyancy. To investigate the boundary layer, a pair of momentum integral equation and heat transport equation is derived considering an elemental volume of fluid on the mountain slope. The governing equations for the core and the boundary layer are then combined, non-dimensionalized, and simplified to be solved numerically.

Prior to numerical treatment, a set of initial conditions are found analytically by considering the behaviour of the wall jet close to the starting point. Comparison is made between our case and a model of an Aluminum Ingot by Davidson, which shares some common features with our flow. Detailed analysis is performed to explain why the solutions differ greatly from the exponential forms found in the latter model, and to suggest an alternative power law relation. The shooting method is then applied where the suitable initial condition is selected according to the new solution forms.

These solutions from numerical integration are presented graphically and magnitudes of key quantities are estimated. While it is understood that turbulent flows are important in the nature, the flow was initially modelled as being laminar until later calculation proved that it is turbulent. Due to the complexity of the problem and the constraint of time, however, we have kept the original model and replaced the laminar viscosity and diffusivity by the turbulent ones.

For a particular slope angle, we have been able to plot the velocity and temperature fields throughout the entire valley. These plots are closely examined in relation to the physics of the flow. In contrary to common boundary layers, the maximum velocities in the valley are found inside the boundary layer. Entrainment of the core flow occurs when the wall jet accelerates over the depth across which core temperature stratification is significant. Heat diffusion slowly homogenises temperature in the core and the thermal boundary layer vanishes as the bottom of the valley becomes isothermal. As a result, the wall jet decelerates and detrains.

The slope angle is then altered to study its effect on the flow. It is found that a steeper slope would result in a greater isothermal region and delay wall jet detrainment.

It is suggested that future work could focus on better turbulence modelling, in particular the correlation terms that were neglected in our laminar model. Also, it would be interesting to investigate and compare with the open ended case where the flow is allowed to exit the plane. Experiments are also recommended to improve the boundary conditions imposed and to verify our theoretical predictions.